4.4 Action Groups Control

This section aims to guide you on how to write and execute the action groups of MiniArm via the Arduino platform.

1. Read Action

1.1 Basic Program Download

The source code of the program is located in: “1.Tutorials\4.Secondary Development Basics and Environment Configuration\4.3 Brief Control Introduction\read_mini_actions\read_mini_actions.ino”.

Note:

◆ Remove the Bluetooth module before downloading the program. Or the program will fail to download because of the serial port conflict.

◆ Please switch the battery box to “OFF” when connecting the Type-B download cable. This action prevents the download cable from accidentally touching the power pins on the expansion board, which may cause a short circuit.

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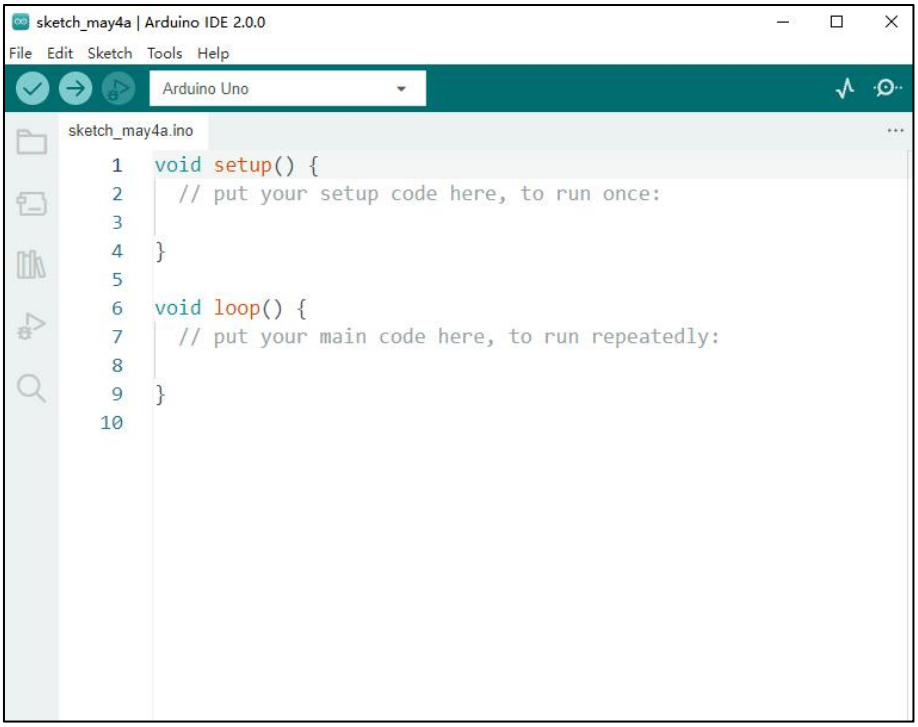
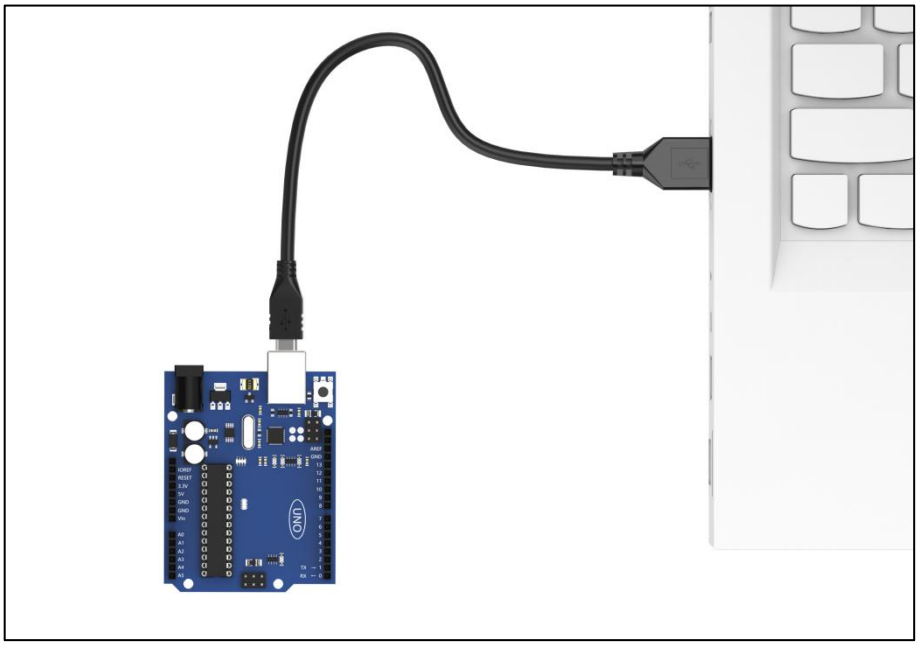
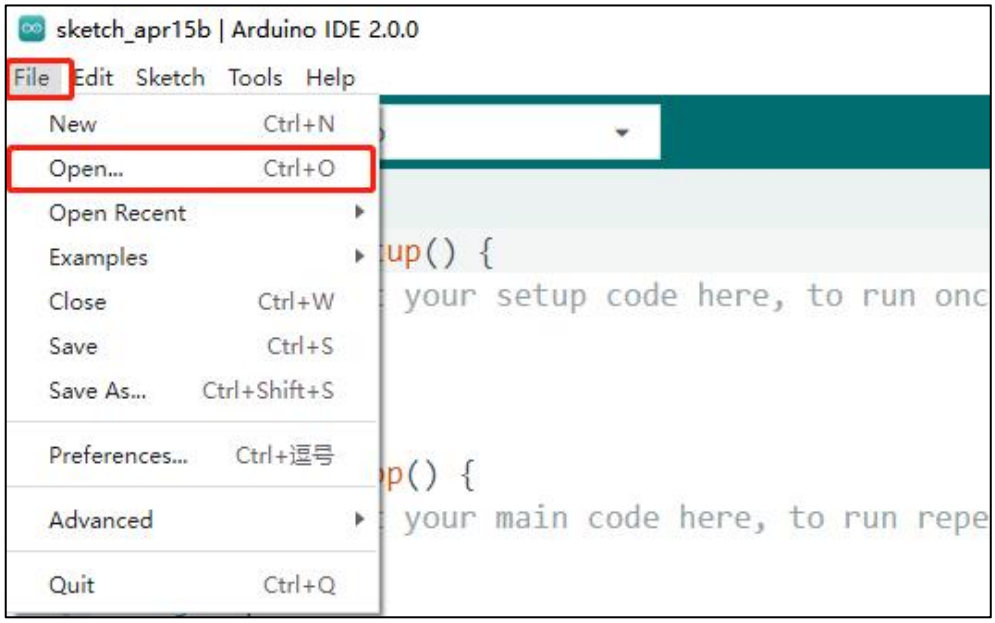
1) Double-click to open it.



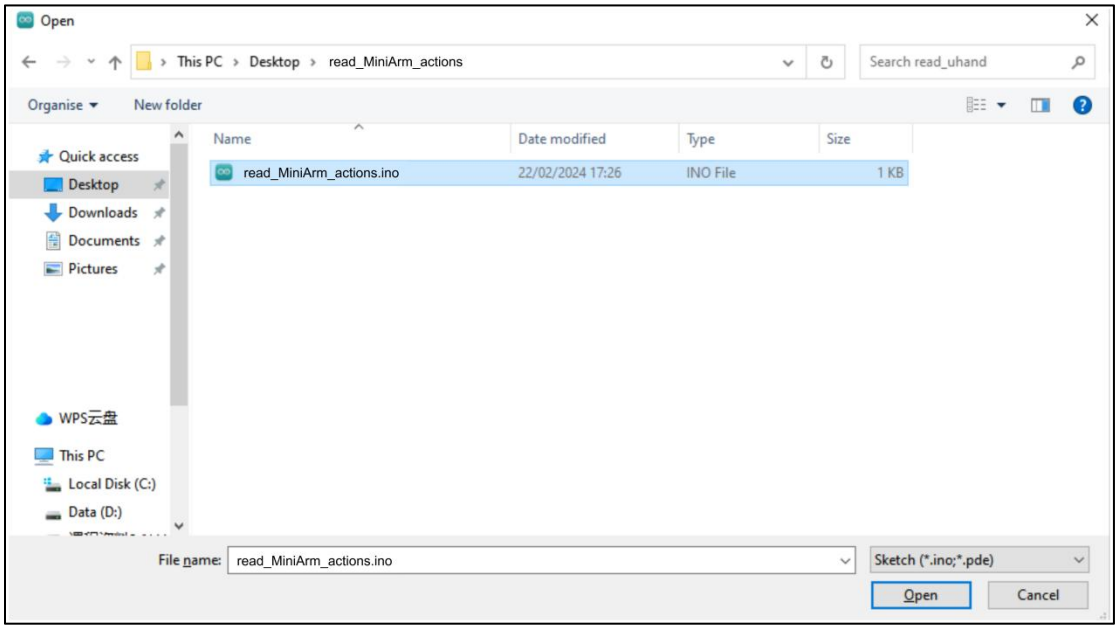
2) Connect the Arduino to the computer with the UNO cable (Type-B).



3) Click “File -> Open”.



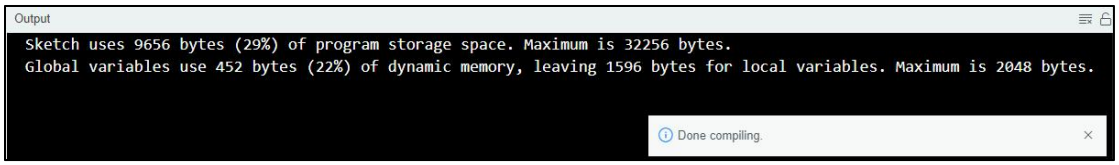
4) Access “1.Tutorials\4.Secondary Development Basics and Environment Configuration\4.3 Brief Control Introduction”. Select the “read_mini_actions.ino” program in the folder “read_mini_actions” and click “Open”.



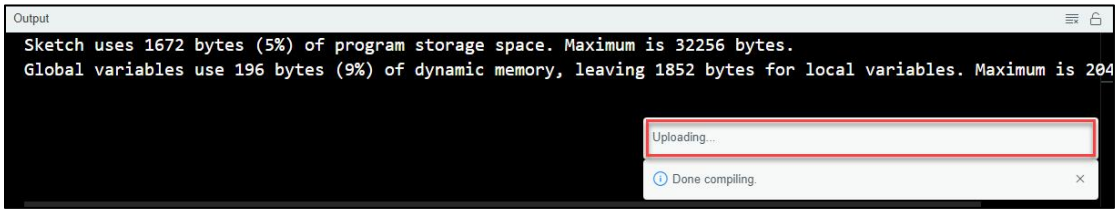
5) Locate the Arduino development board in “Select Development Board”. Take Arduino Uno as an example. COM ports are not unique, and their numbers can be checked in the computer's device management. Take COM6

as an example.

6) After confirming that everything is correct, click to verify the program. If the program is correct, the status area shows “Compiling Project -> Completed” in turn. Then it displays the number of bytes used by the current project, the program storage space, etc.



7) After successful compilation, click to upload the program to the development board. The status area shows “Compiling Project-> Uploading-> Upload successfully” in turn. After that, the status area stops printing and uploading information.



1.2 Get Action Data

1) Please keep the Type-B connected to the computer. Do not plug off the cable during the process to ensure communication.

2) Press “Ctrl+Shift+M” to open the serial port monitor to check the real-time angle data of servos No.1 to No.5. The 0th bit in the data tells whether the data of the current action is valid or not. “1” is valid data used to determine whether the action has been executed successfully during the execution of the

action.

```
Output   Serial Monitor ×
Message (Ctrl + Enter to send message to 'Arduino Uno' on 'COM10')   New Line   9600 baud
{1 ,90 ,0 ,130 ,125 ,90},
{1 ,90 ,0 ,130 ,125 ,90},
{1 ,90 ,0 ,130 ,125 ,90},
{1 ,90 ,0 ,130 ,125 ,90},
{1 ,90 ,0 ,130 ,125 ,90},
{1 ,90 ,0 ,130 ,125 ,90},
```

3) By rotating the knobs S1 to S5 on MiniArm, you can adjust the corresponding angles of servos. Then save the actions in a group to make a new action group.

```
{1 ,74 ,34 ,169 ,135 ,89},
{1 ,74 ,34 ,169 ,135 ,89},
{1 ,90 ,90 ,90 ,90 ,90}
```

2. Action Group Running

2.1 Write Action Group

1) Add the acquired action group data into the file “\mini_actions_group \actions.h” in the same path as this section.

2) The macro definition “action_count” represents the number of current action groups. If a new action group needs to be added, please add 1 to the original data.

```
#include <Arduino.h>
#define action_count 2 //Number of action groups

static uint8_t action[action_count][20][6] =
    {
      //Action Group 1
      {{1 ,90 ,0 ,130 ,125 ,90},{1 ,93 ,0 ,141 ,135 ,0},
      {1 ,93 ,0 ,141 ,135 ,174},{1 ,90 ,90 ,90 ,90 ,90}, {0,0,0,0,0,0}}
      //Action Group 2
      {{1 ,74 ,34 ,169 ,135 ,89},{1 ,74 ,34 ,169 ,135 ,89},
      {1 ,90 ,90 ,90 ,90 ,90}, {0,0,0,0,0,0},
    };
```

When adding the data of action groups, please ensure that the current action data is the two-dimensional array. In action group data, the 0th bit indicates whether the current action group is valid data. “1” indicates that the action is running, and “0” indicates that the action group running is finished. The 1st to 5th bit respectively indicates the corresponding servo angle of No.1 to No.5 servos. Note: Each action group data must end with {0,0,0,0,0,0} to indicate the current action group running has finished.

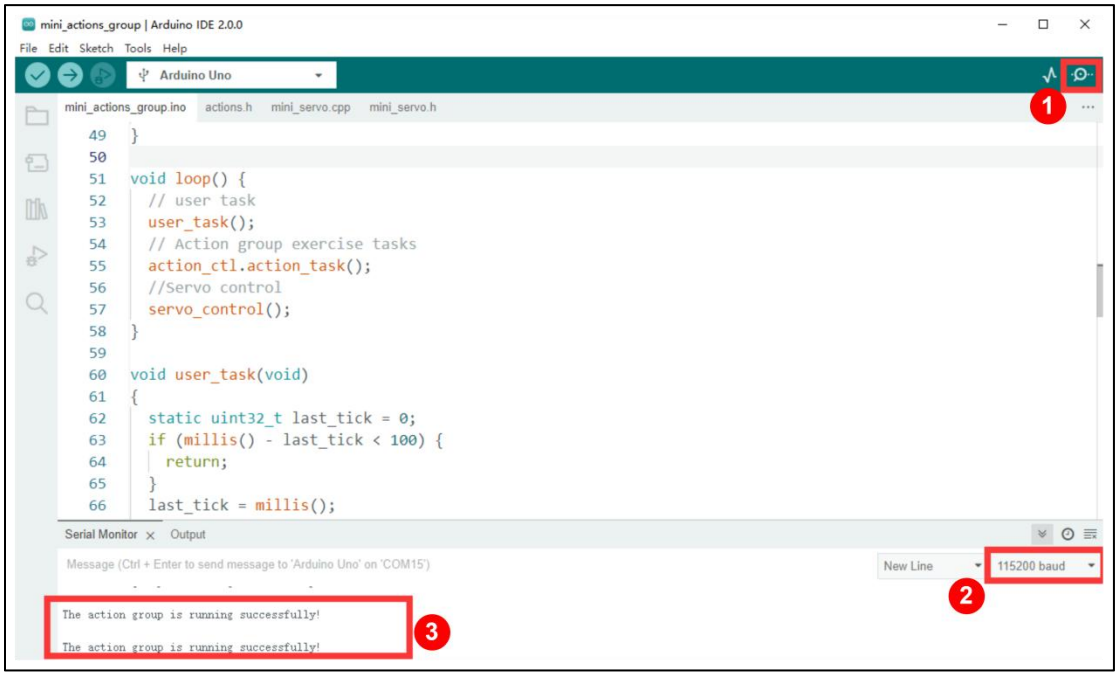
3) After adding the action group data, refer to “1.1 Basic Program Download” above to download the program “mini_actions_group.ino” into Arduino UNO.

4) After the download is complete, power the robotic arm on to run the set action groups.

2.2 Action Group Running Outcome

Power the robotic arm on and keep the Type-B connected to the computer. Open the serial port monitor to select the baud rate as “115200”.

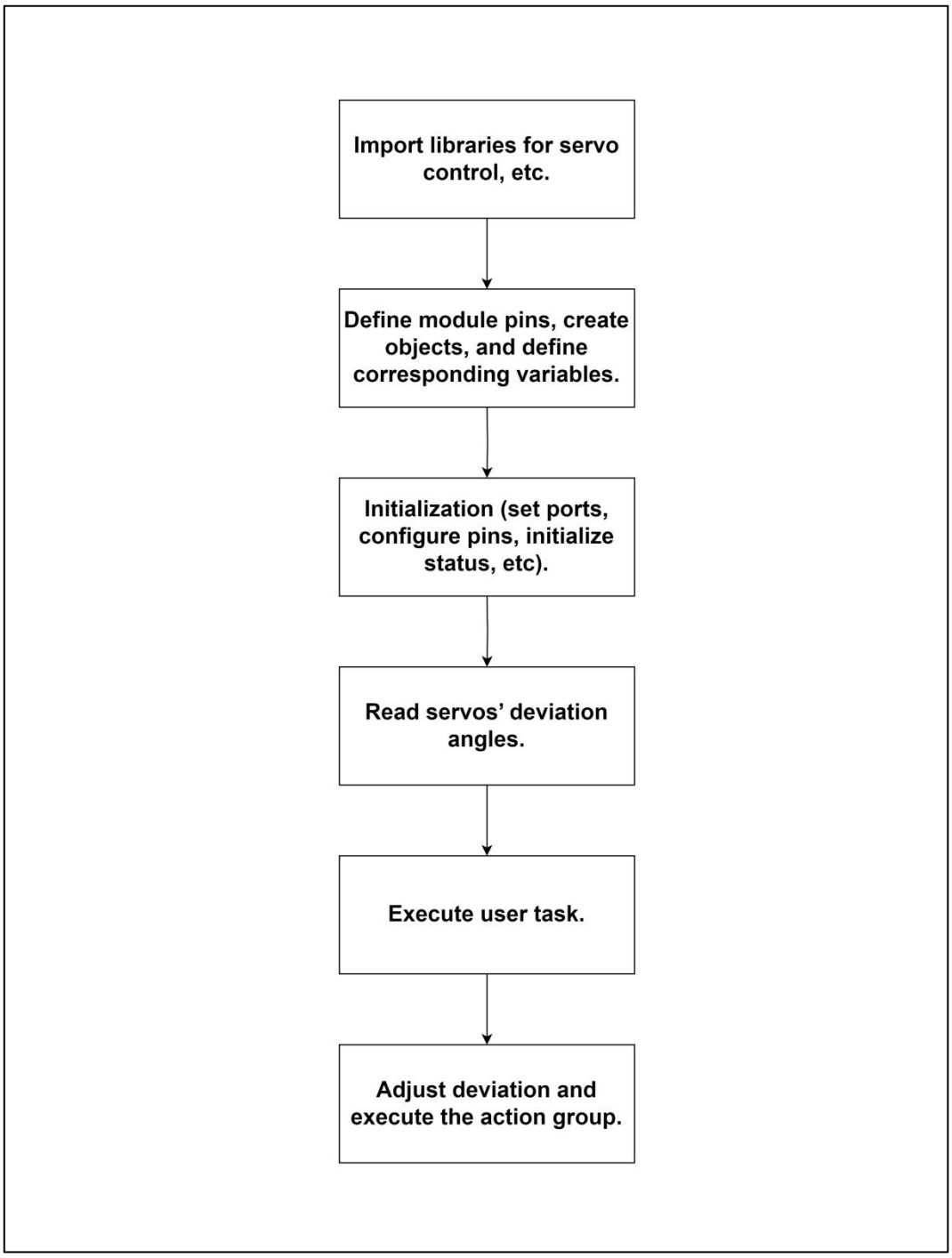
Once the action group is completed, the serial port monitor will display “The action group is running successfully!”, indicating the successful execution of the action group.



2.3 Analysis of Action Group Running Program

The program is located in “1.Tutorials\4.Secondary Development Basics and Environment Configuration\4.3 Brief Control Introduction\ mini_actions_group\ mini_actions_group.ino”.

You can refer to the following flowchart for the implementation logic of the program:



1) Initialize the serial port as "115200", then the servo pins. Call the error reading function "read_servo_offset()" to read the current deviation of the robotic arm.

```
void setup() {
  // put your setup code here, to run once:
  Serial.begin(115200);
  // Set the serial port's timeout for reading data.
  Serial.setTimeout(500);

  // Assign servos' IO port
  for (int i = 0; i < 5; ++i) {
    servos[i].attach(servoPins[i]);
  }

  //Call error reading function
  read_servo_offset();

  delay(2000);
  Serial.println("start");
}
```

2) Within the “user-task ()” function, set the sequence number of the currently executing action group by “action_ctl.action_set()” function. Take the code “action_ctl.action_set(1)” as an example. Set the current action group number to be executed as “1”. Get the currently executing action group through the “action_state_get( )” function, whose return value is the sequence number of the currently executing action group. If the execution is complete, it returns “0”.

```
void user_task(void)
{
  static uint32_t last_tick = 0;
  if (millis() - last_tick < 100) {
    return;
  }
  last_tick = millis();

  static uint32_t step = 0;
  switch(step)
  {
    case 0:
      //Action group control
      action_ctl.action_set(2);//Execute Action Group 1
      Serial.print("action run.");
      step = 1;
      break;
    case 1:
      if(action_ctl.action_state_get() == 0)
      {
        Serial.println("");
        Serial.println("The action group is running successfully!");
      }else{
        Serial.print(" .");
      }
      break;
  }
}
```

3) Use the “action_ctl.action_task()” function to set the corresponding servo angles for the current action.

```
void HW_ACTION_CTL::action_task(void){
  static uint32_t last_tick = 0;
  static uint8_t step = 0;
  static uint8_t num = 0 , delay_count = 0;
  if(action_num != 0 && action_num <= action_count){
    // Interval time
    if (millis() - last_tick < 100) {
      return;
    }
    last_tick = millis();
    switch(step){
      case 0: //Running actions
        if(action[action_num-1][num][0] != 0){
          extended_func_angles[0] = action[action_num-1][num][1];
          extended_func_angles[1] = action[action_num-1][num][2];
          extended_func_angles[2] = action[action_num-1][num][3];
          extended_func_angles[3] = action[action_num-1][num][4];
          extended_func_angles[4] = action[action_num-1][num][5];
          step = 1;
        }else{ //If the operation is completed
          num = 0;
          // Clear action group variables
          action_num = 0;
        }
        break;
```

```
      case 1: //Waiting for action to run
        delay_count++;
        if(delay_count > 2){
          num++;
          delay_count = 0;
          step = 0;
        }
        break;
      default:
        step = 0;
        break;
    }
  }
}
```

4) Next, call “servo_control()” to control servos rotating to corresponding angles for achieving related action. Adjust the deviation of the current robotic arm before running the action.

```
void servo_control(void) {
  static uint32_t last_tick = 0;
  if (millis() - last_tick < 20) {
    return;
  }
  last_tick = millis();

  for (int i = 0; i < 5; ++i) {
    servo_expect[i] = action_ctl.extended_func_angles[i] + servo_offset[i];
    if(servo_angles[i] > servo_expect[i])
    {
      servo_angles[i] = servo_angles[i] * 0.9 + servo_expect[i] * 0.1;
      if(servo_angles[i] < servo_expect[i])
        servo_angles[i] = servo_expect[i];
    }else if(servo_angles[i] < servo_expect[i])
    {
      servo_angles[i] = servo_angles[i] * 0.9 + (servo_expect[i] * 0.1 + 1);
      if(servo_angles[i] > servo_expect[i])
        servo_angles[i] = servo_expect[i];
    }

    servo_angles[i] = servo_angles[i] < limt_angles[i][0] ? limt_angles[i][0] : servo_angles[i];
    servo_angles[i] = servo_angles[i] > limt_angles[i][1] ? limt_angles[i][1] : servo_angles[i];
    servos[i].write(i == 0 || i == 5 ? 180 - servo_angles[i] : servo_angles[i]);
  }
}
```

If modifying the rotation speed of the robotic arm's servos, you should ensure that the sum of actual and desired values is "1".

For instance, if you modify the actual value as "0.5", you should change the corresponding desired value to "0.5". After modifying, download the program, and will see a significant increase in the rotation speed of the robotic arm's servos.